

homework-2

#packages

Question 1

```
install.packages("aRtsy")
```

`install.packages()` is used specifically to install a specific package onto your computer and only needs to be run when installing the package for the first library. The `library()` function's purpose is to load an already existing package into the current session.

Question 2

`library(aRtsy)`

The Collatz conjecture is a famous unsolved math problem that states that if you repeatedly apply the rules of if a number is even: divide it by 2, and if it is odd: you would multiply it by 3 and add 1. If you repeatedly apply these rules you will always get 1.

Collatz conjecture Latex equation:

$$f(n) = \begin{cases} n/2 & \text{if } n \text{ is even} \\ 3n + 1 & \text{if } n \text{ is odd} \end{cases}$$

Question 3:

This function creates a visual representation of the Collatz conjecture. The “n” argument determines the number of starting values, while the color shows what color

part B:

- You get a different output every time because the function uses randomization. Although it is the same code, there are different random values generated, leading to different results
- It would be troublesome to put this code into a .qmd file because the random values of function can be different, leading to varying results, which means the output of the function will look different each time it is rendered
- The `set.seed()` function essentially lets us control R's number generation. By setting a seed, R will consistently produce the same random values each time the code is run.

```
set.seed(13)

canvas_collatz(
  colors = c("blue"),
  n = 10
)
```

